

WORK PROCEDURE

PRO-OP-ITH-004A

PRODUCTION DRILLING ITH-CUBEX

OBJECTIVE

The objective of this procedure is to define safe methods for carrying out production drilling from 3 7/8" to 6 1/2".

APPLICATION

This procedure applies to all MACHINES ROGER INTERNATIONAL employees.

INTERPRETATION

The personnel with authority to interpret this document are the General manager and the Operations Manager. All requests for amendments, revisions or updates must be made to one of these persons.

RESPONSABILITIES

The worker responsible for drilling must ensure that this procedure is applied and that the instructions from the engineering department are followed.

PROCEDURE FOR PRODUCTION DRILLING

1. A daily inspection of the workplace and drilling equipment must be carried out;
2. Signs or barricades are placed at the entrance to the work site to inform all persons around that work is in progress. They must be installed at least 3 metres (10 feet) from the first pieces of equipment or material. See EQUIPMENT POSITIONING STANDARD. Only authorised persons are allowed in the work area.

For UP holes that will subsequently be reamed with the V-30, ensure that the height between the floor and the roof is between 12 feet and a maximum of 14 feet, depending on the drilling angle. It is important to note that the measurement between the slip plate and the tip of the cutters is 7 feet 6 inches.

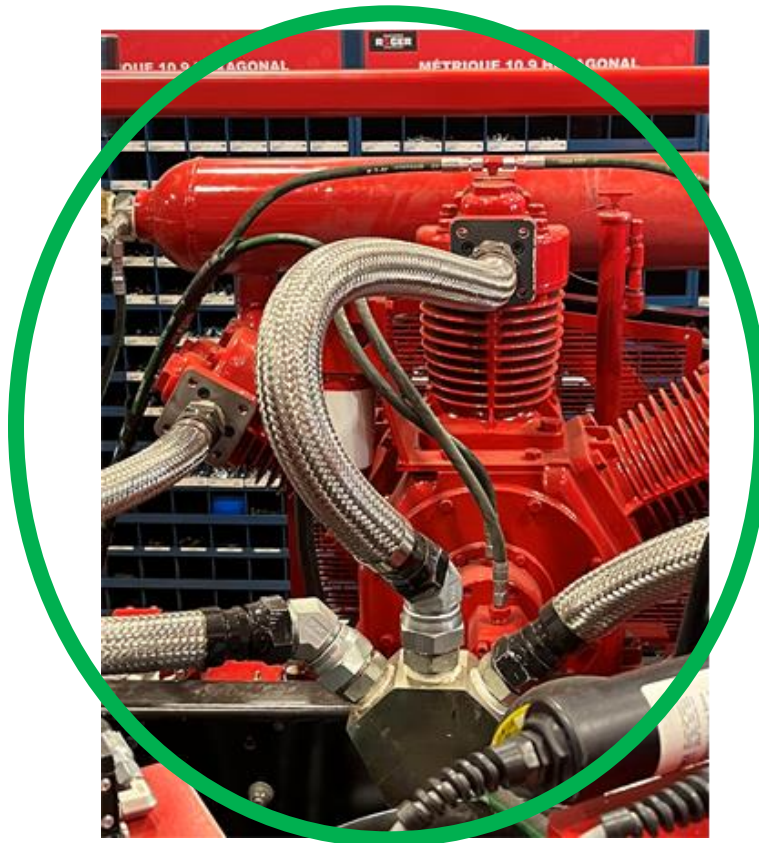
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When the space between the floor and the roof of the work area exceeds 4.3 m (14 ft), the floor of the work area must be raised with excavated rock.

For production drilling, the height of the roof must not allow more than one joint in the void (between the tables and the roof).

3. The electrical cable is unloaded, inspected and connected to the power supply. An electrician is required if the motor rotation needs to be reversed or changed.
4. Make sure that the hydraulic control levers are in neutral, and that they are free from tangles with hoses, electrical wires, etc., before switching the drill "ON".

It is essential to carry out a rigorous inspection of our equipment as soon as it is due to be operated. For CUBEX and boosters, particular attention must be paid to the high temperature (800 degrees F) braided hoses on the compressors. For safety reasons, only the type of hose shown below should be used.



5. When the drill is moved in the drilling area, take the necessary precautions to avoid damaging the electric cable that follows the drill. If the extension needs to be handled while powered up, **use the plastic hook.**

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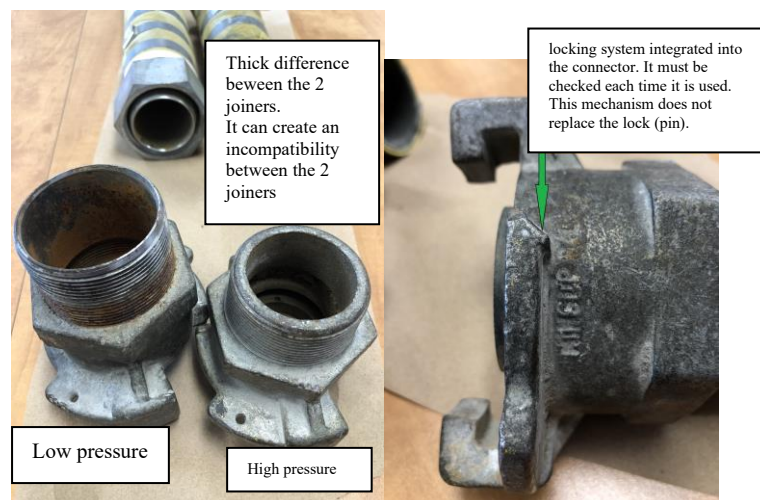
6. Once at the drilling point, switch off the drill by pressing the E-STOP. Hang the electric cable on the wall at chest level (4 to 5 feet from the bench) to eliminate any contact with loose rocks or mining vehicles. A visual inspection of the cable should be carried out during this step to detect any breakage.

USING THE E-STOP TO STOP THE DRILL CUTS THE 600-VOLT CURRENT TO ALL THE EXTENSIONS

7. Connection of the low-pressure air hose (2" dia.) from the mine air supply system to the drill booster.
8. The whip check devices are installed and inspected at each joint (if any). Check that the wire rope is in good condition and that the clamping mechanisms are working properly.
9. The reinforced high-pressure hose (2½") is connected the booster, inspected and blown. Then connect from the booster to the drill rig, installing retainers at each connection. (Note: The stem nuts on the 2½" dia. hoses are designed so that they cannot be interchanged between low and high pressure to eliminate any potential accidents).

Special care required

Couplings and hoses should be inspected thoroughly to ensure that collars are tight, coupling threads are in good condition and hoses are free from defects.



10. The 1" diameter water hose is inspected, blown and connected to the booster and drill.
11. Hang the 600V electric cable on wall

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12. If it is not physically possible to suspend the cable (e.g. the dimensions of the work site) it must be extended in such a way as to eliminate any possible contact with water or equipment. They can be lifted off the bench with wooden cans or tripods and marked (painted or identified with a banner) to be more visible.
13. All hoses are installed to eliminate contact with loose rock and/or mining equipment while maintaining unobstructed access to the work site.
14. Water and air lines are carefully pressurized and inspected for leaks.
15. The drill console must be positioned in such a way as to protect employees from any moving parts or rock projections that may occur when cutting the pilot hole collar.

THERE MUST BE A MINIMUM DISTANCE OF 1.5 METRES BETWEEN THE MAST AND THE CONSOLE. WHEN A PIECE IS IN MOTION, NO WORKER MAY STAND BETWEEN THE CONSOLE AND THE MAST, OR WITHIN 1.5 METRES OF THE MAST ON ANY SIDE.

16. The drill is then positioned over the drilling line (on line and at the desired inclination) as specified in the plans and specifications supplied by the mine, using a magnetic laser and an inclinometer for quality control.

WHEN MOVING THE DRILL OVER A METAL COVER THAT IS ON THE RAISE, IT IS IMPORTANT TO BE CAREFUL TO PREVENT THE COVER FROM GETTING CAUGHT BETWEEN THE TRACTION OF THE DRILL TO AVOID BREAKAGE AND DISPLACEMENT OF THE METAL COVER.

17. When moving the rotating mast, particular attention must be paid to the hydraulic hoses, as they could become jammed. If this happens, go back and move the mast to soften the hoses and loosen them. If effort is required to loosen the hoses, make sure you are in a stable position.
18. If a casing is to be installed, refer to procedure **PRO-OP-ITH-014A Procedure for installation of casing at the beginning of the hole;**
19. The holes must be drilled in accordance with the plans and specifications;
20. **When there is a breakthrough holes, it is important to push, WITHOUT ROTATION, 1 rod more than the length of the breakthrough;**
21. Holes drilled down must be identified and a red cone must be installed at the collar or according to the customer's instructions;
22. To install or remove the drill rods, the gear box must be raised and stopped. The drill must be in the neutral position before the drill rods are handled;

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23. To handle the 6" hammer, the operator must use the transport trolley or the V-30 bell and chain. Make sure the trolley is stable before placing the hammer on it. As a last resort, the hammer could be transported manually with the help of another worker. Changing drill bits and rods can be done alone, depending on the length of the holes drilled.

For holes of 200 feet or more, we prefer to have a 2nd worker on site for drilling and pulled out rods. If it is not possible to have this 2nd worker, the driller must plan breaks.

WARNING
DRILL BITS MUST BE HANDLED WITH CARE. YOU MUST ESTIMATE THE DISTANCES TO REMOVE THEM AND AVOID PINCHED POINTS.

Ajouts ou modifications	Date	Par
Added : moving the CUBEX over the V30 metal cover between points 16 th and 17 th	December 3rd 2020	S. Tremblay
Add point #17 of the possibility that the hydraulic hoses can be stuck and the measures to be taken if this occurs.	12-01-2023	S. Tremblay
Added warning on inspections and high-temperature hoses	7-04-2023	S. Tremblay
Addition of a 3-meter distance between the ribbon and the first pieces of equipment in the work area. Addition made to point #7 and on the " STANDARD INSTALLATIONS" following the Amaruk event.	19-12-2023	S. Tremblay S. Desrosiers
Add at #21 pushing 1 more rod when the break through is done	10 juin 2024	S. Tremblay JF Gagnon
Complete review	March 2025	Operations
Add measurements precisions between point #2 and #3	October 2025	Operations and H&S



Christian St-Amour
Operation Manager

A distance of at least 3 metres (10 feet) must be left between the barricade at the site entrance and the first pieces of equipment (toolbox, oil, etc.).

